

The Impact of the 2016 Brexit Referendum on UK Gilt Yields

A Synthetic Control Method Analysis

Macroeconomics

Introduction

The June 23, 2016, Brexit referendum represented a major political and economic shock to the United Kingdom. Gilt yields, themselves, have become prevalent since discourse about the 'bond market' being a magic hand behind the exchequers actions. We understand debt to be developing into a serious issue for the economy, as are the ramification of Brexit, even ten years on. Counterfactuals in economics are often difficult to identify since no concrete empirical evidence can be identified from a passage of time that did not occur, though we can employ statistical techniques to create a synthetic dataset from which conclusions can be drawn.

Methodology: Synthetic Control Method (SCM)

To accurately isolate the causal effect of the Brexit vote, we employ the **Synthetic Control Method (SCM)**.

Key Concepts:

- **Treatment Event:** The Brexit Referendum (June 23, 2016).
- **Treated Unit:** UK 10-Year Gilt Yields.
- **Synthetic Unit:** A weighted average of equivalent bond yields from a selection of 'donor' countries, selected from a matching algorithm to best mimic pre-Brexit statistics

Let Y_{it} be the true value of the 10 year gult yield of the treated unit (the UK) at month t , and let J be the number of control units in the eligible donor pool in the domain from $j = 2$ to $j = J+1$. A pre-treatment period T_0 is the period prior to the Brexit Referendum.

Further let X_1 be a $K \times 1$ vector of predetermined characteristics of the treated unit in period T_0 , and let X_0 be a $K \times J$ vector of corresponding values for the list of potential donor countries. For clarity and convenience allow $K = 6$ to represent three selected predictor: the policy rate, CPI and GDP growth and three further pre-treatment yield anchors observed in the years before the referendum.

This synthetic control is formalised as a $J \times 1$ vector of weights $W = (w_2, w_3, \dots, w_{J+1})'$ where the weights are non negative and sum to 1. The optimal weight vector, W^* is derived by minimising the distance between the T_0 characteristics of the treatment unit and the synthetic donor pool:

$$W^* = \arg \min_W (X_1 - X_0 W)' V (X_1 - X_0 W)$$

where V is a $K \times K$ vector is a semi definite diagonal matrix that determines the relative importance of each predictor described within V . V is selected to minimise the T_0 RMSPE (Root Mean Square Prediction Error) of the outcome variable.

The post treatment counterfactual is therefore an environment where Brexit did not occur and is defined as a weighted average of the optimal donor pool. Hence the causal effect of the referendum of UK gilt yields is the difference between the synthetic yields and the actual yields.

A selection of robustness checks will follow.

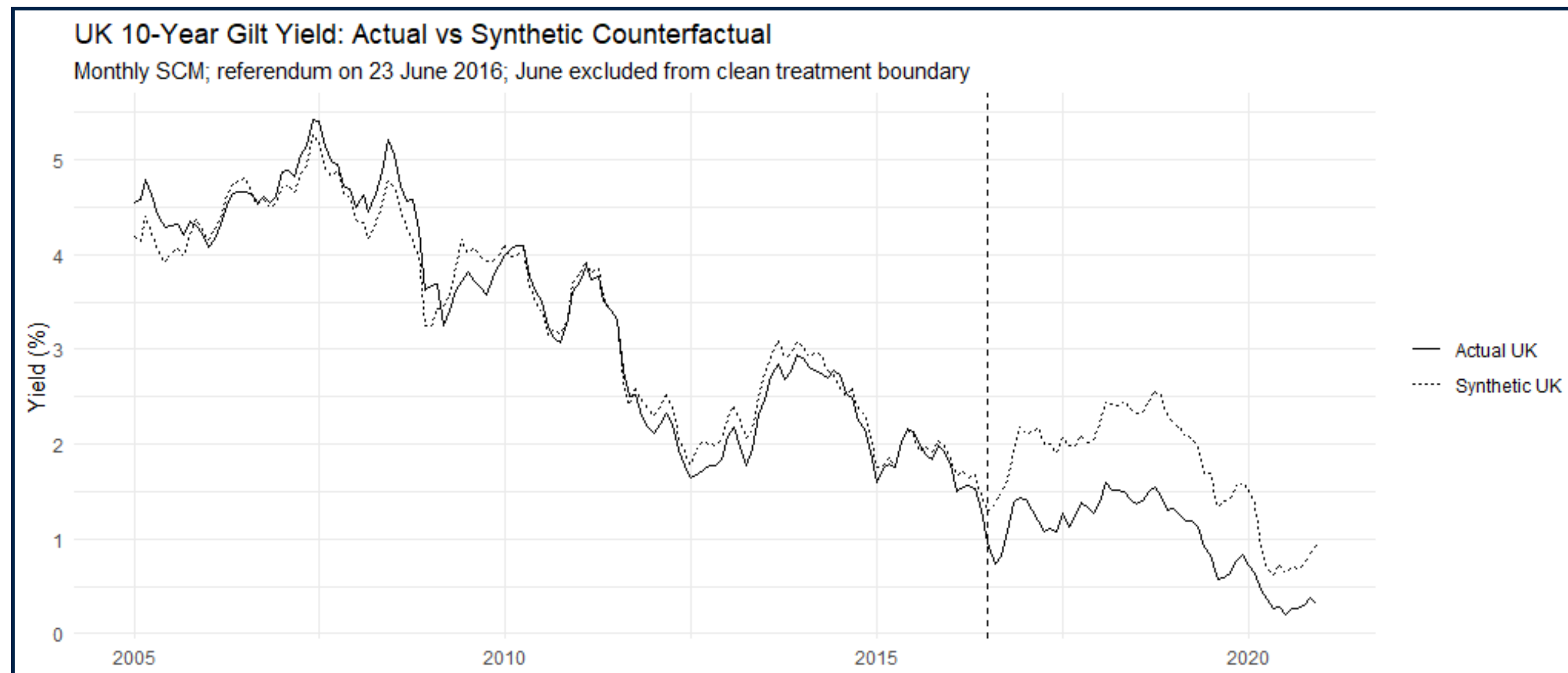
Economic Implications

The sudden drop in actual yields relative to the synthetic baseline is indicative of increasing pessimism of the UK economy:

- **Anticipation of Slower Growth:** New information is quickly priced into the value of the bonds. Agents within the bond market expect slower growth, maybe even recession, which implies lower long term interest rates.
- **Central Bank Intervention:** Investors anticipated that the BoE would lower the baseline policy rate to stimulate a sluggish economy.
- **Flight to Safety:** Political uncertainty drives capital out of risky/productive assets and into risk free assets, pushing bond prices up and yields down.

Results: Yield Divergence

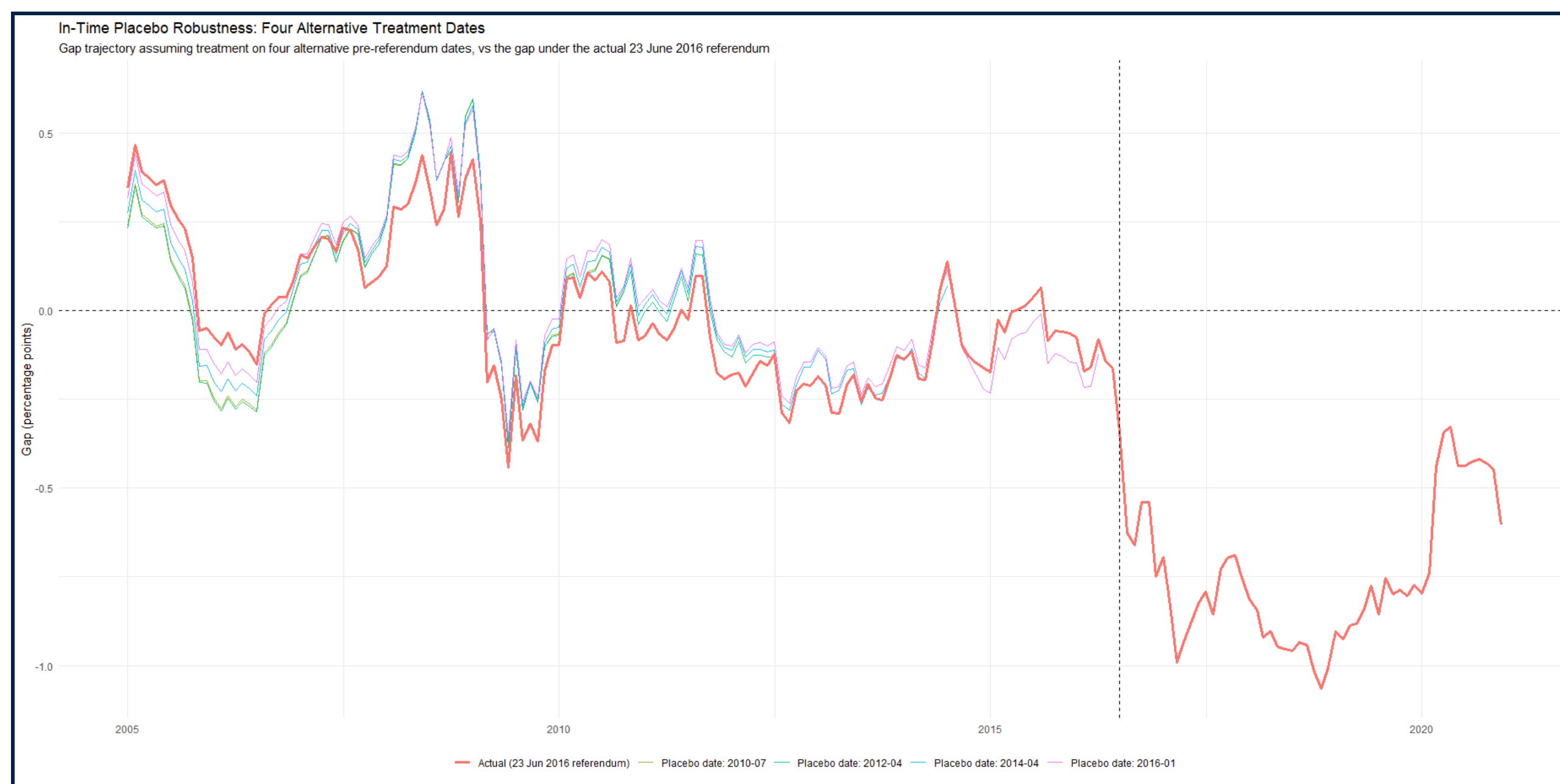
The figure below compares the actual UK 10-year gilt yield against the synthetic counterfactual.



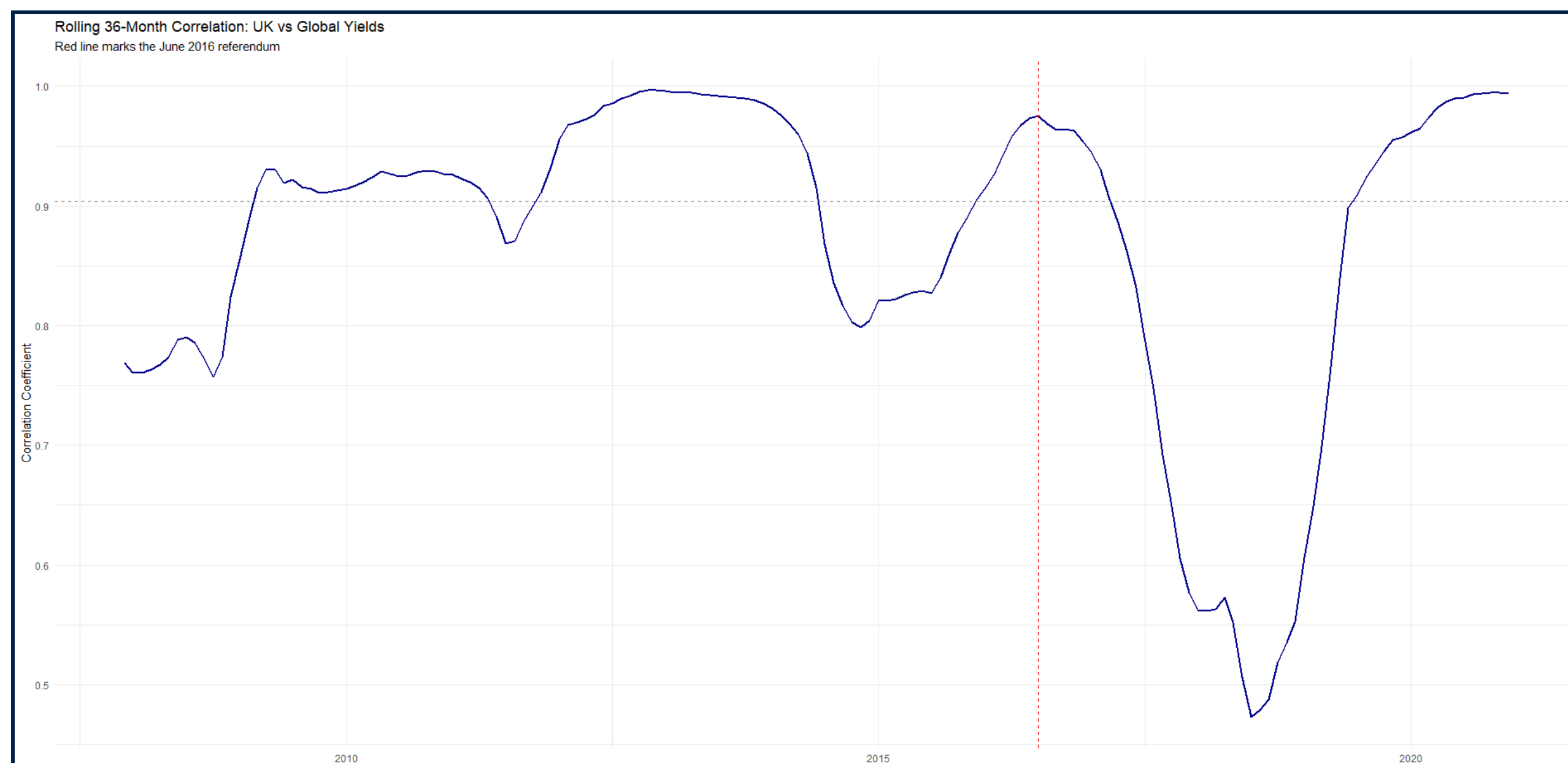
Key Observations:

- **Pre-Treatment Fit:** Prior to June 2016, the Synthetic UK (dotted line) tracks the Actual UK (solid line) with high fidelity, validating the SCM model design.
- **Post-Treatment Divergence:** Immediately following the vertical dashed line (Referendum Date), a sharp divergence occurs.
- **Sustained Impact:** The actual yield drops noticeably below the synthetic estimate and remains suppressed for the subsequent years, highlighting a persistent structural shift in market expectations.

Validation



To provide a holistic argument, we have already discussed the *what*, the *when* now it only requires us to provide the reasoning to the *why*. International Yield co-movement, is a fundamental economic regularity. This means that since economies are intertwined, and economies that are more intertwined tend to follow the same monetary, fiscal and economic policy, as well as exerting convergence effects onto one another; the economies business cycles and economic outlook often appear similar. Although it is not a structural regularity, we can derive significant predictive power about one nations gilt yield by observing its close economic neighbours.



Conclusion

Using the Synthetic Control Method, we demonstrate that the 2016 Brexit referendum exerted a significant downward pressure on UK 10-Year Gilt yields.